



HOW ADANI GROUP'S OPEN ACCESS FUEL FARMS ARE POWERING INDIA'S AVIATION BOOM

As India's aviation sector soars past record passenger numbers, the infrastructure major is quietly reshaping its fuel supply backbone. By pioneering open access fuel farms at major airports, it is helping airlines control costs, enhance safety and future-proof operations against volatile aviation turbine fuel prices

India's aviation sector is taking off at an unprecedented pace. Already the world's third-largest domestic air travel market after the US and China, the country has doubled its operational airports from 74 in 2014 to 157 in 2024. The government's ambitions are higher still — targeting 200 airports by 2025 and 350–400 by 2047. Passenger traffic has surged in step. In the financial year (FY) 2024, Indian airports handled 37.41 crore passengers, a 15 per cent year-on-year rise. Domestic flyers have doubled in a decade and air traffic is expected to touch 30 crore by 2030.

But behind the glossy terminals and growing flight networks lies a stubborn challenge: fuel. Aviation turbine fuel (ATF) accounts for between 30 and 40 per cent of airline operating costs, and its price swings with global crude oil trends. From OPEC+ production cuts to geopolitical conflicts, volatility in ATF prices can wipe out airline profits overnight.

In such a cost-sensitive market, operational efficiency and resilience are critical. That is where open access fuel farms — pioneered at scale by the Adani Group — are transforming Indian aviation.

Beyond cost control, centralised open access fuel farms deliver safety and sustainability gains. Safety is strengthened through international-standard compliance, automated monitoring, and reduced congestion in high-risk airside zones.

Breaking the Old Model

Traditionally, Indian airports relied on closed-access fuel farms run by a single supplier. This system curtails competition, thereby diminishing the bargaining power of airlines.

Open access fuel farms flip the script. They create shared infrastructure where multiple suppliers can store and dispense ATF, bringing in competition, better pricing and supply flexibility. For airlines, this means a crucial hedge against global oil shocks due to prevailing geopolitical tensions.

The Adani Group, through Adani Airport Holdings Ltd. (AAHL), is leading this transition. Of India's 12 open access fuel farms, seven are at Adani-operated airports: Mumbai, Lucknow, Ahmedabad, Guwahati, Thiruvananthapuram, Jaipur and Mangaluru. The upcoming Navi Mumbai International Airport (NMIA) is also being designed on this model.

Hydrant Refuelling System

One of the key features of the Open Access model is that it paves the path for a Hydrant Refuelling System, which otherwise is difficult to have in a closed access model. Modern fuel farms are the energy backbone of an airport. Large storage tanks receive ATF, which is then piped directly to aircraft via hydrant refuelling systems. Earlier, most airports depended on bowsters — tanker trucks ferrying fuel to parked planes. While effective, this meant constant vehicle movement on the airside, raising safety risks and slowing aircraft turnaround times.

Thiruvananthapuram airport gets future ready

Thiruvananthapuram International Airport (TRV) recently commissioned its open access fuel farm and aircraft refuelling facility.

The facility has been developed through the strategic consolidation of existing infrastructure from multiple oil marketing companies operating at the airport. Over the next 24 months, Thiruvananthapuram will also commission a modern jet fuel storage system equipped with a hydrant refuelling system, enabling faster, safer, and more efficient aircraft refuelling.



Hydrant systems change the game. With underground pipelines connecting storage to aircraft stands, refuelling becomes faster, safer and more reliable.

Defending Against ATF Volatility

ATF volatility is one of the biggest risks for Indian carriers. In recent years, global disruptions — from the ongoing Russia–Ukraine conflict to OPEC+ cuts — have sent fuel prices soaring.

Open access fuel farms provide a structural safeguard. By enabling multiple suppliers to operate in the same hub, they allow airlines to diversify procurement and